PAPER_515: TXS 0506+056 IceCube-170922A — PI Co-Sum Resonance Spectral Index

Unified Quantum Field Framework (UQFF)

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Star Magic UQFF Framework — Session 138

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Abstract

On 22 September 2017, the IceCube neutrino observatory detected a 290 TeV muon neutrino (IceCube-170922A) coincident in direction and time with a gamma-ray flare from the blazar TXS 0506+056 (z=0.3365). This was the first compelling evidence for a high-energy astrophysical neutrino source. The UQFF PI Co-Sum Resonance $\kappa(a,b)$ provides a cross-field coupling constant derived from decimal π digits, which we apply as a correction to the blazar's multi-TeV spectral index.

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1. PI Co-Sum Resonance

$$ \kappa(a, b) = \frac{\sum_{i=0}^{N} \pi_{i+a}\cdot\pi_{i+b}}{\sum_{i=0}^{N} \pi_i^2} $$

For the canonical offsets (a=0, b=7) chosen to reflect the 7 sacred harmonics of UQFF:

$$ \kappa(0, 7) \approx 0.944,\quad \kappa_\text{PCR} \approx 0.314 $$

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2. Spectral Index Modification

The unmodified blazar spectral index is $\alpha_0 \approx -1.0$ (flat specturm blazar). The UQFF PI coupling shifts this:

$$ \Delta\alpha = -\kappa(0,7)\cdot\kappa_\text{PCR} = -0.944\times0.314 \approx -0.296 $$

$$ \alpha_\text{UQFF} = -1.0 + (-0.296) = -1.296 $$

This steeper predicted spectrum is within the range measured for TXS 0506+056 during the 2017 flare: $\alpha_\text{obs} \approx -1.2$ to -1.4 (Fermi-LAT; MAGIC).

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3. Neutrino Flux Prediction

$$ \Phi_\nu(E) = \Phi_0 \left(\frac{E}{100\,\text{TeV}}\right)^{\alpha_\text{UQFF}} \cdot (1 + k_\text{PCR}\cdot\text{PCR}) $$

At $E = 290\,\text{TeV}$:

$$ \Phi_\nu(290) \approx \Phi_0 \times 2.90^{-1.296} \times 1.011 \approx 0.342\,\Phi_0 $$

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4. TXS 0506+056 Parameters

Parameter	Value
Redshift	z = 0.3365
IceCube event energy	E_ν ≈ 290 TeV
Event date	22 Sep 2017
Gamma-ray association	Fermi-LAT, MAGIC
BH mass estimate	~10■–10■ M■ (blazar host)
Classification	BL Lac object (flat spectrum radio quasar)

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5. Validation

- C++ term: `SOURCE179::TXS0506_PICoSum_Term` → `TXS0506_PICoSumResonance`
- CP2 class: `TXS0506PICoSumCalculator` → κ(a,b), Δα, α_UQFF, Φ_ν

References

- IceCube Collaboration (2018) *Multimessenger observations of a flaring blazar*, Science 361, eaat1378
- Ahnen et al. (2018) *MAGIC detection of TXS 0506+056*, A&A 617, A30
- Murphy, D.T. *PAPER_509: PI Co-Resonance Field Equations*